

CRISTOFF STAN

Senior Software Engineer

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PROFESSIONAL SUMMARY

Innovative software engineer with over 9 years of experience in building scalable, high-performance backend systems. Passionate about low-level systems engineering, deterministic execution, and blockchain infrastructure. Seeking to contribute deep Rust expertise and distributed systems insight to a cutting-edge blockchain platform.

TECHNICAL SKILLS

- **Languages:** Rust (advanced), Go, TypeScript, Python, C#, JavaScript, SQL
- **Systems/Blockchain:** Solana (runtime concepts), SPL, Anchor, Ethereum, Protocol Buffers, gRPC
- **Distributed Systems:** Kafka, Redis, gRPC, WebSockets, Event-Driven Architectures
- **Infrastructure:** Docker, Kubernetes, NGINX, Terraform, GitHub Actions, AWS, Azure, Linux
- **Databases:** PostgreSQL, MongoDB, MySQL, Redis, SQLite
- **Tools/Frameworks:** Actix, Rocket, Tokio, Axum, Anchor, React, Next.js, FastAPI, .NET Core
- **Testing & CI/CD:** Unit Testing, Integration Testing, GitHub Actions, Docker Compose, Playwright

WORK EXPERIENCE

Senior Software Engineer

Sensidev

September 2021 – December 2024

Timișoara, Romania (Remote)

- Led design and implementation of performance-critical backend services using Rust and Go for decentralized analytics tools across multiple cloud environments.
- Developed fault-tolerant services with gRPC, Kafka, and Redis to support event-driven pipelines processing millions of records per hour in near real-time.
- Wrote deterministic, low-latency compute modules and transaction engines inspired by Solana-like execution models for high-throughput workflows.
- Implemented network orchestration logic and modular protocol runners with Tokio, ensuring horizontal scalability and resilience under load.
- Introduced a secure sandbox execution layer for contract-like logic in Rust, enabling custom extensions by enterprise clients.
- Architected and maintained a microservice mesh using Kubernetes, implementing automated deployment pipelines with GitHub Actions and Helm.
- Developed blockchain indexing utilities to stream and validate chain data via WebSockets and HTTP/JSON-RPC, laying groundwork for integration with L1/L2 chains.

- Conducted rigorous code reviews with a focus on modularity, memory safety, and concurrency correctness, raising the overall codebase reliability.

Software Engineer

Syscom Digital

October 2019 – August 2021

Bucharest, Romania (Remote)

- Engineered backend APIs for IoT telemetry ingestion and command orchestration using Go and TypeScript, managing over 50K concurrent edge devices.
- Designed distributed logging and alerting infrastructure for anomaly detection pipelines using ELK Stack and Rust-based telemetry processors.
- Delivered core infrastructure modules for a modular integration system, allowing seamless onboarding of external services and APIs via sandboxed Rust runners.
- Led a cross-functional effort to containerize legacy services and enable cloud-agnostic deployment on Azure and AWS via Docker and Terraform.
- Maintained and extended the platform's extensibility layer using secure, sandboxed plugin architecture for third-party script execution.
- Designed schema-first APIs with GraphQL and integrated domain event handlers with Kafka and Webhooks for decoupled processing.
- Developed low-overhead rate-limiting and tracing middleware for internal APIs using tower-layered Axum services.
- Mentored junior engineers on topics including concurrent design patterns, protocol development, and cloud-native architecture.

Software Developer

Soft Build

June 2016 – July 2019

Timișoara, Romania (Remote)

- Built resilient cloud-first microservices handling financial data aggregation, using Rust and C# for high-performance ETL pipelines.
- Implemented fault-isolated service runners using Tokio for asynchronous job scheduling and distributed task orchestration.
- Authored core protocol handlers and serialization logic for custom internal RPC protocol inspired by gRPC, focusing on deterministic behavior.
- Developed real-time dashboards using WebSocket feeds and data sync mechanisms, providing insights into cross-system operations.
- Conducted load testing and fine-tuned memory allocation patterns in Rust services to meet real-time latency constraints.
- Contributed to internal SDKs for message signing, verification, and identity provisioning within secured enclaves.
- Designed and documented internal P2P messaging protocol over QUIC for low-latency, loss-tolerant communication between services.
- Helped introduce CI/CD pipelines and Git branching strategies that improved delivery speed and code quality.

Software Developer
Decima Digital
November 2015 – April 2016
Tallinn, Estonia

- Developed performance-sensitive backend features in C# and Rust prototypes for a blockchain data verification module.
- Collaborated on research involving consensus protocols and DAG-based data synchronization models.
- Wrote simulation tools to test edge cases in distributed peer discovery mechanisms and message broadcasting.
- Contributed to dynamic module loader built in Rust to allow hot-reloading of backend service logic.
- Integrated blockchain data parsing engines with PostgreSQL and Redis for high-speed indexing and analytics.
- Deployed core services in Docker containers on early Kubernetes setups, automating service discovery with Consul.
- Instrumented test coverage tooling and static analysis for low-level modules to ensure secure memory usage.
- Documented internal architecture and presented findings on event-driven communication models and consensus strategies.

Software Developer Intern
Borna Technology
September 2014 – August 2015
Tallinn, Estonia (On-Site)

- Developed backend modules using Rust and Python to parse and store telemetry from a fleet of mobile sensors.
- Built an embedded rules engine to analyze environmental data using decision trees and basic machine learning models.
- Integrated PostgreSQL with spatial indexing extensions (PostGIS) to support geolocation-based alert systems.
- Built RESTful APIs in Flask and Rocket for mobile clients to sync collected data and receive live updates.
- Designed and tested a basic peer-to-peer file transfer system over LAN, using low-level socket programming in Rust.
- Developed CLI tools for data export, batch processing, and schema migrations using Diesel ORM.
- Implemented caching logic and TTL-based memory stores to improve system response latency.
- Contributed to the company's first Docker adoption initiative by writing containerization guides and Dockerfiles.

EDUCATION

Tallinn University

Bachelor's Degree in Computer Science,
2010 – 2014

CERTIFICATIONS

- ❖ Rust Programming Masterclass – Udemy
 - ❖ Distributed Systems & Scalability – Coursera (University of Illinois)
 - ❖ Blockchain Architecture – Consensys Academy
 - ❖ Kubernetes for Developers – CNCF
 - ❖ System Programming in Rust – Second State Academy
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PROJECTS

Solana-Lite Runtime Prototype

- Designed a simplified transaction execution runtime in Rust inspired by Solana's account model, focusing on deterministic state updates and parallel execution.
- Integrated simulated transaction pool, runtime verifier, and storage abstraction layer.

Rust Event Engine for Distributed IoT Systems

- Created a publish-subscribe event engine in Rust that managed data from 10K+ sensors with sub-second latency using Tokio and async channels.

Distributed Key-Value Store with Raft Consensus

- Built a fault-tolerant distributed key-value store using Raft, written in Rust with full persistence and replication support.